

AT-FSV / ATAI

CMFSB ISOLATION MODULE — DOCUMENT 00

MASTER ENGINEERING BASELINE | ATFSV-MOD-001

Captured Mechanical Floating Safety Body

Revision 5.18 · Contract Diligence Baseline · Licence & IP-Transaction Negotiation

Issue date: 2026-10-06 · Legal Owner / Author: Amirbahador Atai

Field	Controlled value
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Classification	T1 — Confidential / NDA controlled. Annex T3-A is T3 and shall be removed from any T1/T2 distribution
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Approval authority	TBD after independent review
Next revision trigger	ICD-001 draft · SIM-001 numerical run · STR-001 stop curve · ACT-001 actuator map · ENE-001 envelope · packaging residual plan · IP filing-status update · reviewer disposition

Controlled limitation. This document supports T1 technical diligence, supplier evaluation, Path A licence negotiation and IP-transaction discussion. It is not a performance certificate, type approval, safety case, measured ride-comfort claim, production commitment, warranty, patent grant, price list or regulatory approval. R5.18 is a single-layer controlled baseline: no duplicated body, no blank pages, no attitude/pyramid option in the core baseline. Engineering OPEN items remain OPEN until evidence closes them. Factual differences from prior revisions are recorded in a separate project Change Log.

0. REVISION CONTROL AND DEFECT CLOSURE

0.1 Identity rule. One document ID identifies exactly one content set. All earlier files carrying IDs R5.11 through R5.17 are WITHDRAWN and are not active design inputs. Only values in Section 3 (Master Parameter Register) and requirement text in Section 7 are active.

0.2 Defects closed in R5.18 (document-control only — not engineering verification).

#	Defect (from R5.16 / R5.17 review)	Corrective action in R5.18
1	Full body content duplicated (28 pages from 14-page logical document)	Single continuous content layer; one END marker
2	Fragmented internal “Challenge topic” list mid-document	Complete Open Items Register published under §13.4
3	Page numbering inconsistent with file (footer “of 14” on 28-page file)	Footer shows sequential page number only; single pass generation
4	Sparse / near-blank pages	No blank pages; every page carries substantive tables or prose
5	LOCKED geometry readable as packaging closure	§4 states explicitly: LOCKED = engagement target only; residual may be negative
6	“Applied” corrective text readable as engineering closure	§0.3 and C-table separate “Text applied” from “Engineering status”
7	Patent status and buyer-distribution posture ambiguous	§16.5 states “None filed” honestly; provisional preferred before broad T1 distribution

0.3 “Text applied” versus engineering status. A row marked “Text applied” means the controlled wording is present in this document. It does not mean the underlying engineering item is verified, tested, or closed. Engineering status remains OPEN until the named evidence artefact exists.

0.4 Corrective-item register (traceability).

ID	Topic	Section	Text applied	Engineering status
C-01	LOCKED geometry is not packaging closure	§4	Yes	OPEN — V-L residual ≈ -120 mm
C-02	6.3 kW is Lorentz copper loss only	§5	Yes	OPEN — ENE-001
C-03	48 V current budget	§5	Yes	OPEN — ENE-001 / ACT-001
C-04	REQ-M07 quantitative settle criteria	§7	Yes	OPEN — limits not frozen
C-05	REQ-M15 under power and hydraulic loss	§7	Yes	OPEN — physical trial
C-06	REQ-M21 sensor / fault / transient policy	§7	Yes	OPEN — SIM/HIL
C-07	Sensor functional independence	§7 / §9	Yes	OPEN — SAF-001

ID	Topic	Section	Text applied	Engineering status
C-08	Km evidence chain	\$10	Yes	OPEN — ACT-001 map
C-09	Gap tolerance envelope	\$10	Yes	OPEN — ACT-001
C-10	Progressive-stop curve	\$7 / \$12	Yes	OPEN — STR-001
C-11	REQ-M25 horizontal path	\$7	Yes	OPEN — STR-001 / ICD-001
C-12	Multi-family IP, ownership, disclosure	\$16, T3-A	Yes	OPEN — filing status
C-13	Tower mass statistical basis	\$6	Yes	OPEN — supplier data
C-14	Static floor rule and traceability	\$6.2	Yes	Applied; mass inputs STUDY
C-15	Stack-up arithmetic	\$4	Yes	OPEN — residual explicit
C-16	State-machine completeness (S3 entry, S4 exit)	\$8	Yes	Text complete; verification OPEN
C-17	Structural duplication / broken challenge page	This R5.18	Yes	Closed at document level

1. PURPOSE, NON-CLAIMS, VALUE AND DISCLOSURE TIERS

1.1 Purpose. This baseline defines the licensable CMFSB isolation module (ATFSV-MOD-001) for Path A engagement and supports Path B host studies and Path C IP discussion under counsel. It freezes architecture language, safety invariants, dual-mode behaviour, controlled requirements, verification planning and disclosure tiers for T1 diligence.

1.2 Non-claims register (binding for external communication).

#	Non-claim
1	No production readiness or series-production commitment
2	No type approval, certification or regulatory compliance of module or vehicle
3	No certified ASIL assignment
4	No measured ride-comfort percentage or guaranteed comfort improvement until L1/L2 configuration-managed evidence
5	No crashworthiness or survivability demonstration
6	No guaranteed 7,000 kg vehicle compliance from the module alone
7	No final supplier mass, cost or lead-time commitment
8	No clinical outcome claim for ambulance / Application A use
9	No continuous public-road high-speed capability claim
10	No closed mass or energy budget suitable for production release
11	No patent grant, novelty, enforceability or freedom-to-operate conclusion before counsel-led analysis
12	No FIA homologation or series eligibility from FIA-referenced design philosophy
13	No supplier or OEM endorsement
14	No continuous RMS force capability until thermal bench data exist
15	No warranty, price list or offer of exclusive sale by this document

1.3 Value statement and reference application.

Occupant compartments in specialty, defence-utility, ambulance and extreme limited-series vehicles receive road vibration that degrades comfort, task performance and, where relevant, clinical stability. Soft isolation alone can reduce handling stiffness in rapid manoeuvres. CMFSB addresses both by keeping the upper body mechanically captured at all times, providing a COMFORT isolation envelope and a SPORT mechanical lock for handling, with passive support on power loss. Application A (ambulance / clinical transport) is the reference use context for comfort-related requirements; no clinical outcome is claimed (Non-claim 8). Dual running gear (selectable COMFORT isolation versus SPORT lock) is the product differentiator. Module Path A licensing does not require the Application A host; Path B vehicle programmes use the locked hosts in §14.

1.4 Planning KPIs (method definitions only — not measured claims).

KPI	Metric definition	Method	Reference	Status
Ride isolation	a _w (0.5–80 Hz) + VDV; MSDV (0.1–0.5 Hz) per ISO 2631-1	SIM-001 then bench	ISO 2631-1; ISO 8608	OPEN until L1
Handling stiffness	Roll angle / lateral response SPORT vs COMFORT under same step	SIM-001 + mule	Defined lateral proxy	OPEN until L1
Mode time	SPORT↔COMFORT complete transition	Bench / HIL	≤3 s TARGET class	TARGET
Power-loss	Settle on passive support within envelope; valves fail-closed	Power-cut test	S4 invariant	BASELINE rule
Module mass	Core tower ≤55 kg TARGET; full corner STUDY	Weighing	\$6 module table	TARGET / STUDY

1.5 Disclosure tiers T0 / T1 / T2 / T3.

Tier	Content	Access condition
T0	Public-facing non-enabling concept / teaser material	Public or first contact
T1	This confidential engineering baseline: architecture, requirements, risk, verification planning	NDA
T2	Models, detailed ICD, test data, configuration-controlled design artefacts for detailed design and verification	NDA + need-to-know
T3	Patent claim charts, prior-art mapping, negotiation strategy, commercial terms (COM-001), counsel-controlled disclosure	NDA + counsel + restricted list

Commercial conditions (Option validity, exclusivity, fees, credit) are T3 content in COM-001 only. T1 holds a one-line reference to COM-001. Commercial conditions are not part of the T2 definition.

1.6 Status vocabulary (exclusive — use only these labels).

Label	Meaning
BASELINE	Architectural decision locked for this revision; still requires verification
TARGET	Numeric goal for design; not yet verified by test
STUDY	Indicative figure for planning; subject to change
ASSUMPTION	Input taken without programme-owned evidence
DERIVED	Calculated from stated inputs; inherits uncertainty of inputs
OPEN / TBD	Unresolved or not responsibly fixed at this revision
OPTION	Commercial or technical option; not committed in the baseline
LOCKED	Programme engagement decision for geometry/mass; not test verification and not packaging closure
SUPERSEDED / HISTORICAL	Prior numerical content retained for traceability; not valid for current design decisions

2. ARCHITECTURE, SAFETY INVARIANTS AND MODES

2.1 Architecture. The upper body is kinematically constrained at all times (no free levitation). Four corner towers each combine: (1) a hydraulic macro stage (PARAM-M02-MACRO-STROKE) with leveling and lock seats; (2) a Lorentz micro stage (PARAM-M03-MICRO-STROKE) within the PARAM-LORENTZ-GAP envelope; (3) progressive mechanical stops; and (4) an optional SPORT positive lock. A shared ECU implements mode arbitration, the REQ-M21 interlock and power-loss detection. Layer stack, top to bottom: upper occupant/safety body → micro stage → macro stage → lower chassis. Architectural signature: the body is always mechanically captured; active isolation operates inside a bounded mechanical safety envelope; comfort control, handling stiffness and crash containment are coordinated but independently bounded. This baseline does not include an attitude / counter-lean / pyramidal underbody option; any such feature, if pursued later, shall be a separate configuration-controlled option with its own evidence plan.

2.2 Safety invariants (non-negotiable behavioural rules).

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture
Static support	Vehicle weight on passive or fail-safe compliant paths — never the sole magnetic path
Power loss (S4)	S4 passive support; leveling valves fail-closed; software aliveness not required
Crash path	Progressive stops + structure; independent of closed-loop magnetic force
Mode transition	SPORT→COMFORT never enters unconstrained free-float
SPORT lock fault	Lock failure shall not defeat capture or progressive stops
Horizontal restraint	Longitudinal / lateral / yaw loads resolved by dedicated paths and/or SPORT lock (REQ-M25)
Override limit	Driver override cannot defeat the lateral-g interlock (REQ-M21)
Manual rescue	Documented manual rescue release (REQ-M15) remains available via a deliberate two-step action

2.3 Dual-mode definition.

Mode	State ID	Mechanical state	Objective
COMFORT	S2	Locks released; macro envelope active; micro active; leveling active	Soft ride isolation
SPORT	S2S	Upper body seated; all required locks confirmed; isolation stages held	Handling stiffness (roll / yaw)

Lateral-g interlock (REQ-M21 — temporary TARGET freeze). Any SPORT→COMFORT transition (automatic or manual) is inhibited when $|a_y|$ exceeds PARAM-M21-AY-HIGH (0.35 g), with no time filter. Re-enable is permitted below PARAM-M21-AY-LOW (0.30 g) hysteresis. A driver override may queue a request but cannot defeat the interlock. COMFORT maximum steady-state roll gradient at design payload (V-L): ≤ 6.0 deg/g TARGET (SIM-001 / HIL). A separate documented

manual rescue release is defined under REQ-M15 for emergency and maintenance use only.

2.4 Hold function (definition for REQ-M16).

Hold is an optional secondary rate-limited restraint, distinct from the SPORT positive lock and from progressive stops. Implementation class: hydraulic orifice and/or friction device at the macro stage. Engagement is rate-limited (velocity and force limits OPEN — freeze in ICD). Hold is not a primary crash energy path; progressive stops remain ultimate. Mass of hold hardware is included in the corner hydraulic/stop share. If hold is not fitted, REQ-M16 is marked N/A for that configuration.

2.5 Horizontal restraint (longitudinal / lateral / yaw).

Macro travel and progressive stops are specified primarily in the vertical axis. Braking, cornering and crash introduce horizontal loads. Load paths: (1) capture frame and linkage shear capacity; (2) dedicated horizontal stops or links with defined free play; (3) SPORT lock faces providing additional shear/roll restraint when engaged. REQ-M25 owns numeric free play, stiffness and load share. OPEN until STR-001 and ICD freeze geometry. Risk R-15 tracks horizontal-restraint closure. Per-class horizontal magnitudes are given in §6.3.

3. MASTER PARAMETER REGISTER

Only values in this section and requirement text in Section 7 are active design inputs. Status labels are exclusive per §1.6. Where a value is ASSUMPTION, STUDY or OPEN, it shall not be presented externally as verified performance.

Parameter ID	Value	Status
PARAM-M02-MACRO-STROKE	±50 mm body-relative (100 mm total range)	TARGET
PARAM-M03-MICRO-STROKE	±7.5 mm	TARGET
PARAM-LORENTZ-GAP	2–5 mm coil-magnet baseline; 15–30 mm only for PM+EM OPTION	TARGET
PARAM-FORCE-PEAK	±2.5 kN study per tower	STUDY
PARAM-FORCE-RMS	≈0.7 kN RMS planning per tower	STUDY
PARAM-KM	≥63 N/vW ASSUMPTION for 2–5 mm gap — requires FEA / supplier map	ASSUMPTION
PARAM-COPPER-MEAN	≤0.5 kW continuous class at 0.7 kN RMS (four towers)	TARGET
PARAM-COPPER-PEAK	≈1.57 kW per tower; ≈6.3 kW four towers — copper loss only	DERIVED
PARAM-SUPPLY-NOM	48 V class nominal	STUDY
PARAM-BW-CONTROL	≥20 Hz; carriage first structural mode ≥3× control bandwidth	TARGET
PARAM-K-EFF	≈40 kN/m about the leveled operating point	STUDY
PARAM-FN-STUDY	≈1.23 Hz at corner mass share of occupied reference	DERIVED / STUDY
PARAM-REF-MASS-OCC	≈2,650 kg occupied upper-body reference	STUDY
PARAM-REF-MASS-MAX	≈3,150 kg maximum-study upper body	STUDY
PARAM-TOWER-MASS-P50	48 kg core tower	STUDY
PARAM-TOWER-MASS-LIMIT	≤55 kg TARGET at the declared confidence level	TARGET
PARAM-MASS-CONFIDENCE	TBD in CFG-001 before G1 (candidate P90)	OPEN
PARAM-TOWER-MASS-P90-LINESUM	67 kg (conservative line-sum of element P90 values)	STUDY
PARAM-STATIC-RULE	Floor = round-up to 0.5 kN of (1.25 × maximum static corner load)	BASELINE
PARAM-STR-S-STATIC-FLOOR	≥5.0 kN per corner	TARGET
PARAM-STR-M-STATIC-FLOOR	≥7.0 kN per corner	TARGET
PARAM-STR-L-STATIC-FLOOR	≥10.0 kN per corner	TARGET
PARAM-M21-AY-HIGH	0.35 g (SPORT→COMFORT inhibit)	Temporary TARGET
PARAM-M21-AY-LOW	0.30 g (re-enable hysteresis)	Temporary TARGET
PARAM-M21-ROLL-GRAD	≤6.0 deg/g COMFORT at design payload (V-L)	TARGET
PARAM-MODE-TIME	≤3 s SPORT↔COMFORT TARGET class	TARGET
PARAM-LIFE	≥10,000 h equivalent road duty	TARGET
PARAM-REVIEWER-DEADLINE	Earlier of 2026-12-05 and G1 closure	BASELINE process

4. PACKAGING FEASIBILITY GATE

Packaging rule (Defect 5 closure). LOCKED geometry values in §14 are programme engagement targets only. They are not packaging closure. Until H-point, cabin interior height, tower integration, macro travel, free play, stop stroke, micro-stage clearance and chassis integration are configuration-controlled under CFG-001, residual margin may be negative. Do not force a non-negative residual by reducing cabin height below H-point requirements. A negative residual is a P0 open item, not a hidden assumption.

Element	V-L (mm)	V-H (mm)	Status
Overall height (SPORT)	1,700	2,000	LOCKED — engagement target
Ground clearance (SPORT)	200	300	LOCKED — engagement target
Available for structure (derived)	1,500	1,700	DERIVED
Chassis / axle structure	280	320	STUDY — integrate tower where possible
Tower structure + mounts	120	140	STUDY — partial integration in chassis
Macro travel (± 50 mm) + free play 40 mm	140	140	TARGET + STUDY
Cabin interior height from H-point	$\geq 1,000$	$\geq 1,000$	TARGET minimum
Roof / exterior shell	80	90	STUDY
Allocated sum (with $\geq 1,000$ cabin)	$\geq 1,620$	$\geq 1,690$	DERIVED
Residual margin vs available	≤ -120	≤ -10 to $+10$	OPEN — P0; may be negative

V-L residual ≈ -120 mm is an explicit OPEN packaging item (C-01 / C-15). Closure paths under configuration control: (a) integrate tower structure and travel within chassis depth; (b) raise overall height; (c) reduce ground clearance; (d) package free play inside cell underfloor. Residual shall be re-reported at each revision until ≥ 0 under CFG-001. V-H residual is near zero with the stated STUDY numbers; any stop stroke above the residual makes it negative unless absorbed inside the tower allowance.

5. POWER AND CURRENT ENVELOPE

5.1 Rule. PARAM-COPPER-PEAK is Lorentz copper loss at simultaneous peak force on four towers. It is not module electrical demand. Electrical input power equals copper loss plus mechanical power plus driver/converter losses. Total system power shall include: Lorentz actuators; hydraulic pump; valve actuation; lock actuation; ECU and sensors. Final current budget at 48 V is OPEN under ENE-001 / ACT-001 (C-02, C-03).

Consumer	Mean (W)	Peak (W)	Owner document
Lorentz, four towers	≤ 500	$\approx 6,300$ (copper only)	ACT-001
Hydraulic pump	OPEN	OPEN	HYD-001
Valves (leveling + hold)	OPEN	OPEN	HYD-001
Lock actuators	OPEN	OPEN	LCK-001
ECU and sensors	OPEN	OPEN	ICD-001
Total system	OPEN	OPEN	ENE-001

Peak copper ≈ 6.3 kW class (four towers at 2.5 kN). Bus current shall be derived in ACT-001 from the force constant and driver efficiency. Continuous public-road RMS capability is not claimed until thermal bench data exist.

6. MASS, STATIC FLOORS AND LOAD CASES

6.1 Module mass breakdown (licensable object — Class L basis).

The ≤ 55 kg TARGET applies to the core tower only (structure + Lorentz + sensors + local electronics). Full corner assembly ≈ 94 kg is Class L basis. Full corner and shared hydraulics are separate STUDY lines so the licensed module is traceable in the vehicle WBS. Class M corner mass is STUDY (interpolated) until supplier data exist.

Element	P50 kg	P90 kg	Scope
Local structure (per tower)	18	25	Core
Lorentz + local carriage	22	30	Core
Sensors + local electronics	8	12	Core
Core tower subtotal	48	67	≤ 55 kg TARGET at P50; P90 under R-01
Cylinder + accumulator share	15	22	Corner
Valves / hoses / seals	5	8	Corner
Progressive stops	8	12	Corner
SPORT lock share	6	10	Corner
Harness / dampers / bellows	4	6	Corner
Pump + reservoir share (1/4)	8	12	Shared
Full corner assembly (Class L basis)	≈ 94	≈ 137	STUDY — Class L ≈ 95 kg
Four corners total (Class L)	≈ 376	≈ 548	Maps to vehicle WBS isolation line

6.1a Per-class corner mass (planning).

Class	Mapped host	Corner P50 (kg)	4-corner isolation (P50)	Notes
S	V-L (4-seat)	≈70	≈280	Lighter hydraulic share
M	Intermediate	≈85	≈340	STUDY (interpolated) until supplier data
L	V-H / heavy	≈95	≈380	Class L basis; heavier hydraulic / structure

6.2 Static support floor — single rule, traceable inputs.

Rule (PARAM-STATIC-RULE): floor = round-up to the next 0.5 kN of ($1.25 \times$ maximum static corner load). Mass basis for each class is stated so every floor is auditable. Occupant policy is unified at 100 kg per occupant.

Class	Mass basis	Max static / corner	$1.25 \times \text{max}$	Floor (0.5 kN round-up)
S (V-L)	Occupied corner share 310–350 kg	≈4.0 kN (STUDY)	5.0	≥5.0 kN
M (intermediate)	Corner share 450–550 kg	≈5.5 kN (STUDY)	6.88	≥7.0 kN
L (V-H / generic)	PARAM-REF-MASS-MAX $\div 4 \times 9.81$	≈7.7 kN (7.73)	9.66	≥10.0 kN

6.3 Reference load cases (STUDY — set static floor and stop rating).

Case	Class S	Class M	Class L (2,650 kg ref)	Status
Corner mass share (approx)	≈310–350 kg	≈450–550 kg	≈662.5 kg	STUDY
Static / tower	≈2.5–4.0 kN	≈4–5.5 kN	≈6.5–7.7 kN	STUDY
Static floor	≥5 kN	≥7 kN	≥10 kN	TARGET
3 g operational / tower	≈8–12 kN	≈12–17 kN	≈20–23 kN	STUDY
20 g vertical / tower	≈60–70 kN	≈90–110 kN	≈130 kN	STUDY
$\times 2$ asymmetry / tower	≈120–140 kN	≈180–220 kN	≈260 kN	STUDY → STR-001
Horizontal 1 g total (upper)	≈12–14 kN	≈18–22 kN	≈26 kN	REQ-M25
Horizontal 20 g total (upper)	≈250–280 kN	≈360–440 kN	≈520 kN	REQ-M25
Stop energy class	STUDY	STUDY	≈15.6 kJ @ 2,650 kg, $v \approx 3.43$ m/s	STR-001

Stop energy illustration (occupied upper body ≈2,650 kg, free play to stop $s = 30$ mm, $a \approx 20$ g class): $v = \sqrt{(2as)} \approx 3.43$ m/s; $E \approx \frac{1}{2} \cdot m \cdot v^2 \approx 15.6$ kJ. Until STR-001 freezes the force-displacement curve, all energy figures remain STUDY. Risk R-08 compares Class L crash loads (≈130/260 kN) to capture-frame, stops and hardpoint capacity ($L=4$, $I=5$, score 20).

7. CONTROLLED REQUIREMENTS (REQ-M01-REQ-M25)

Requirements below are the controlled module set. Gate column indicates the earliest gate at which objective evidence is expected. Status OPEN means the numeric limit or procedure is not yet frozen — not that the requirement is optional.

ID	Requirement	Owner	Gate	Status
REQ-M01	Upper body mechanically captured in all states; static capacity per class S/M/L at the §6.2 floors	Structures	G0/G2	BASELINE / TARGET
REQ-M02	Macro stroke PARAM-M02-MACRO-STROKE without clash	Mechanical	G1	TARGET
REQ-M03	Micro stroke PARAM-M03-MICRO-STROKE inside PARAM-LORENTZ-GAP	Actuator	G1	TARGET
REQ-M04	Control bandwidth and carriage first mode per PARAM-BW-CONTROL	Control	G2	TARGET
REQ-M05	Peak and RMS force and Km per parameter register	Actuator	G2	STUDY
REQ-M06	Core tower mass per PARAM-TOWER-MASS-LIMIT at declared confidence	Module	G2	TARGET
REQ-M07	Power-loss → S4 passive; valves fail-closed; quantitative settle envelope OPEN until frozen	Safety	G2	BASELINE; limits OPEN
REQ-M08	Multi-band SIM-001 acceptance per §12	Sim	G1	TARGET
REQ-M09a	Progressive-stop screening curve from per-class loads (STR-001)	Structures	G3	TARGET

ID	Requirement	Owner	Gate	Status
REQ-M09b	Stops absorb crash energy without dependence on the magnet force loop	Structures	G3	TARGET
REQ-M10	Life $\geq 10,000$ h equivalent road duty	Reliability	G3	TARGET
REQ-M11	Operating temperature range frozen	Thermal	G2	OPEN
REQ-M12	EMI / cabin field plan	EMC	G3	OPEN
REQ-M13	ICD mech/elec/comm/safety/lock released	Systems	G1	TARGET
REQ-M14	No single failure causes loss of capture	Safety	G2	TARGET
REQ-M15	Manual recovery / rescue access (two-step), including under power and hydraulic loss (trial OPEN)	HF/Safety	G3	TARGET
REQ-M16	If hold fitted: rate-limited orifice/friction; not primary crash path (§2.4)	Safety	G2	BASELINE optional
REQ-M17	Required SPORT locks positively confirm engagement and release; time per PARAM-MODE-TIME	Lock	G2	TARGET
REQ-M18	COMFORT provides full macro envelope without lock interference	Mechanical	G2	TARGET
REQ-M19	Mode arbitration driver/radar/GPS with override hierarchy	Controls	G3	STUDY
REQ-M20	Transition never unconstrained free-float	Safety	G2	BASELINE
REQ-M21	SPORT $\leftrightarrow$ COMFORT lateral-g policy and roll gradient (§2.3; parameters in §3)	Dynamics	G2	TARGET
REQ-M22	Leveling preserves dynamic stroke under payload change; $k \approx 40$ kN/m is rate about leveled point	Hydraulics	G2	TARGET
REQ-M23	Incomplete lock set inhibits SPORT declaration; lock state diagnosable	Lock/Controls	G2	TARGET
REQ-M24	Fluid systems observe ISO 4413 safety rules path	Hydraulics	G1	TARGET
REQ-M25	Horizontal X/Y/yaw free play, stiffness, stop loads and SPORT lock shear share frozen	Structures	G2	OPEN

Leveling note (REQ-M22). PARAM-K-EFF ≈ 40 kN/m is the effective rate about the leveled operating point. Static sag is taken by gas-spring preload and active leveling, not by the ± 50 mm dynamic stroke. A payload change $\Delta F \approx 2.4$ kN would give ≈ 60 mm at 40 kN/m; therefore active leveling is required to preserve dynamic stroke.

8. STATE MACHINE (S0-S8)

State names below are authoritative. SPORT mode corresponds to state S2S. COMFORT corresponds to S2.

State	Name	Rule
S0	Safe / Captured / Power-off	No powered release; mechanical capture remains
S1	Precharge / Self-test	Capture OK; 48 V path checked
S2	COMFORT isolation	Full envelope; capture mandatory
S2S	SPORT locked	All required locks confirmed; stages held
S3	Degraded	Reduced authority; force bounded
S4	Power-loss capture	Passive support; valves fail-closed
S5	Impact preparation	Prepare stops; inhibit high-rate unlock
S6	Impact / arrest	Stops active; no magnet-energy dependence
S7	Post-event safe	No automatic re-enable
S8	Maintenance lockout	Physical lockout engaged

8.1 Transition table (complete).

From	To	Condition / authority
S0	S1	Power-on self-test pass
S1	S0	Shutdown command (controlled power-off)
S1	S3	Self-test fail or critical fault detected
S1	S2	Driver or automatic mode select (COMFORT)
S1 / S2 / S2S	S8	Physical lockout engaged; only when stationary
S2	S2S	SPORT request; all required locks report engaged

From	To	Condition / authority
S2S	S2	SPORT→COMFORT allowed by REQ-M21; locks report released
S2 / S2S	S3	Degraded capability detected (fault class per ICD)
S2 / S2S	S5	Impact-preparation trigger: pre-crash sensor class and/or deceleration threshold (OPEN until ICD freeze)
S3	S1	Fault cleared + authorised reset
S3	S4	Power loss
S4	S1	Power restored + self-test pass (only path out of S4)
S5	S6	Measured impact / stop contact
S5	S2 / S2S	Impact-preparation abort
S5	S4	Power loss
S6	S7	Event handling complete → post-event safe
S7	S1	Manual clear by authorised personnel only
S8	S1	Maintenance lockout removed by physical key / procedure
Any (power loss)	S4	Valves fail-closed; locks spring-applied fail-locked if engaged

S5 entry trigger class (pre-crash sensor / deceleration threshold) remains OPEN until ICD freeze and is tracked under the Open Items Register. S2/S2S→S3 is defined for degraded capability; verification remains OPEN (C-16).

9. SAFETY ANALYSIS (FMEA AND PRELIMINARY HARA)

Ratings below are PRELIMINARY screening labels for programme planning. They are not a formal ISO 26262 ASIL assignment. Formal HARA, FTA and safety case work products are owned by SAF-001 at G2.

9.1 Summary FMEA.

Failure mode	Effect	Mitigation	S	O	D
Carriage jam	Loss of micro authority	Sensors; stops; passive support	9	3	4
Link fracture	Local free-motion risk	Capture frame; stops	9	2	5
Leveling stuck	Stroke consumed	Level sensor; S3	8	3	3
Hose burst	Loss of support / leveling	Pressure sense; stops	9	3	3
Coil overheat	Derate / shutdown	Thermal model	7	3	3
Magnet Km loss	Force-constant drop	Temp limits; derate	7	2	4
SPORT lock fail to engage	Remain in COMFORT	Status sensors; driver alert	6	3	2
Partial lock (1-2 of 4)	Asymmetric stiffness	Inhibit S2S declaration	8	2	2
False COMFORT at high lat-g	Excess roll	Lat-g / yaw inhibit (REQ-M21)	7	2	3
Horizontal path overload	Capture shear exceedance	REQ-M25; structural stops	9	2	4
Power loss	Active force lost	S4 invariant (fail-closed)	—	—	—
Software / OTA fault	Bad command	Bounds; S3 / S4	8	3	4
Unintended SPORT lock at speed on rough road	Handling / comfort conflict	Arbitration inhibit; alert; R-17	8	3	3
Accumulator pre-charge loss	Macro response degraded	Pressure sense; S3; R-18	7	3	3
Hydraulic leak / fire	Loss of support; fire risk	Fluid selection; containment; R-19	9	2	4
Common-cause sensor fault	False mode or force	Diverse sensing; cross-check; R-20	8	3	3

Failure mode	Effect	Mitigation	S	O	D
Magnet demagnetisation	Force-constant drop	Temp limits; derate; R-21	7	2	4

9.2 Preliminary HARA (screening).

Hazard	Scenario	S/E/C cand.	Safety goal (draft)
Loss of capture	Carriage or link failure	S3/E3/C3	Maintain capture under single failure
Uncontrolled vertical motion	Force runaway	S2/E3/C2	Bound force; enter S3 or S4
Asymmetric SPORT lock	Partial lock + lateral load	S2/E3/C2	Inhibit S2S unless lock set complete
Horizontal overload	Brake/crash without sized path	S3/E2/C3	REQ-M25 paths sized in STR-001
Unexpected height drop	Valve open at speed	S2/E3/C2	Fail-closed valves
Post-crash auto re-enable	Uncommanded motion after event	S2/E2/C2	S7 holds until manual clear
Unintended SPORT lock at speed on rough road	False SPORT engage	S2/E3/C2	Inhibit; alert; R-17

Process candidates: ISO 26262 (E/E elements of isolation control and mode arbitration); ISO 21448 (SOTIF); ISO 13849 (lock/valve safety functions if used); UN R100 / ISO 6469-3 (HV safety if applicable); UN R94/R95/R14/R16 (crash/belts if applicable); ISO 2631-1 for ride metrics; EN 1789 / EN 1865 for Application A planning only.

10. ACTUATOR MAP, GAP ENVELOPE AND THERMAL

Quantity	Value	Comment
Km	≥63 N/√W	ASSUMPTION until FEA / supplier map for the 2-5 mm gap
RMS planning force	≈0.7 kN per tower	Aligned with ≤0.5 kW mean copper (four towers)
Peak study force	±2.5 kN per tower	Pulse thermal closed in ICD / ACT-001
Peak copper (4 × 2.5 kN)	≈6.3 kW class; bus current per ACT-001	Driver and wiring sizing
Gap baseline	2-5 mm coil-magnet	15-30 mm applies only to PM+EM OPTION
SPORT continuous copper	Near zero	Stages held; locks carry static

Km evidence chain and gap tolerance envelope are OPEN under ACT-001 (C-08, C-09). Demagnetisation risk is tracked as R-21. Continuous public-road RMS capability is not claimed until thermal bench data exist.

11. MODE SEQUENCES AND LOCK ARCHITECTURE

11.1 COMFORT → SPORT sequence (TARGET ≤3 s under defined conditions).

Step	Action	Interlock
1	Mode request accepted by arbitration	No active inhibit (lat-g, fault)
2	Macro / leveling drives upper body to seat band	Stroke sensors in range
3	Seat contact detected at required corners	Contact flags true
4	Positive locks command engage	Actuator enable healthy
5	All required locks report engaged	Incomplete set → abort to COMFORT
6	Isolation stages held; state S2S (SPORT)	Driver indication ON

Forbidden: declaring SPORT while any required lock remains disengaged.

11.2 SPORT → COMFORT sequence.

Release is inhibited above the lateral-g threshold (REQ-M21): any SPORT→COMFORT transition is inhibited at |a_y| > 0.35 g, no time filter; re-enable below 0.30 g. Driver override cannot defeat this interlock. When allowed: (1) locks command release; (2) all report released; (3) macro envelope re-enabled; (4) micro control re-enabled; (5) state S2 (COMFORT). Forbidden: declaring COMFORT while any required lock remains engaged. Manual rescue release (REQ-M15) is a separate documented two-step procedure for emergency and maintenance only.

11.3 Lock architecture planning (ATFSV-LCK-001).

Element	Function	Status
Seat faces	Upper hard-point to lower receiver at seat band	Geometry OPEN — ICD
Positive lock	Pin, claw or hydraulic clamp — prevents separation	Per-corner status required
Actuation	Hydraulic or electromechanical; may share macro power	Duty cycle STUDY
Fail behaviour	Fail to known state with driver alert; never free-float	BASELINE rule
Crash role	May share load; progressive stops remain primary energy path	BASELINE

11.4 Horizontal load path summary (Class L reference).

Load	Order of magnitude (Class L / 2,650 kg)	Path
Brake 1 g	≈26 kN total	Capture shear + links + SPORT lock if engaged
Lateral 1 g	≈26 kN total	Same; roll couples into vertical towers
Crash 20 g horizontal class	≈520 kN total	Structure + horizontal stops; not Lorentz
Vertical crash with 1.5-2 asymmetry	≈260 kN per tower class	Stops + cell; R-08

Per-class horizontal magnitudes for S and M are in §6.3. Numeric free play and stiffness freeze under REQ-M25 / STR-001.

12. VERIFICATION FRAMEWORK

Item	Planning definition
Model class P0	Quarter-car / half-car: upper mass, macro k & c, optional micro force, SPORT = high-k constraint
Reference passive	Frozen passive hydro-pneumatic k & c stated on every comparative plot
Road class	ISO 8608 segments; at least one rough and one smooth profile
Mass states	Empty / occupied / maximum as §6
Frequency bands	a_w (0.5–80 Hz) + VDV; MSDV (0.1–0.5 Hz) per ISO 2631-1
COMFORT policy	Lateral-g policy and roll-gradient (≤6.0 deg/g) per REQ-M21
Outputs	Transmissibility; band RMS; stroke histogram; roll proxy SPORT vs COMFORT; transition time history
SIM-C1	COMFORT lower RMS in ≥1 agreed band vs frozen passive on rough profile — TARGET after valid run
SIM-C2	SPORT higher roll stiffness / lower roll than COMFORT on same lateral step; roll metric defined in SIM-001 — TARGET
SIM-C3	Macro within ±50 mm without continuous stop hammering on agreed rough profile
SIM-C4	No unconstrained free-float interval in any transition case
Evidence ladder	L0 architecture (this DOC) → L1 SIM-001 → L2 single-tower bench → L3 four-corner / HIL → L4 mule optional

SIM-001 primary outputs: frequency-weighted RMS a_w (0.5–80 Hz), VDV, and MSDV (0.1–0.5 Hz) per ISO 2631-1. Band-limited tables may be retained as supplementary diagnostics only. Results marked PRELIMINARY until independent review.

12.1 L2 / L3 test outlines.

L2 single-tower: force-gap-current curve; power-loss landing to S4; lock engage/release within PARAM-MODE-TIME; thermal RMS; contamination robustness; stop contact within STR-001 curve. L3 four-tower / HIL: corner coherence including asymmetry; mode transition; power-loss policy from each mode; fault injection; controlled stop engagement without loss of capture. Every evidence record retains software version, parameter set, configuration hash, road/profile ID, payload, mode, passive k/c, time histories, uncertainty where applicable, and review status.

13. GATES, PHASES, RISKS, WORK PACKAGES AND OPEN ITEMS

13.1 Gates G0-G5.

Gate	Intent	Entry evidence (planning)
G0	Concept controlled	DOC-00 issued; legal owner recorded
G1	Interface and SIM ready	ICD draft; SIM assumptions frozen; clash check; packaging residual plan
G2	Single-tower evidence	Force, thermal, power-loss, lock bench; ACT-001 map start; REQ-M07 limits frozen
G3	Four-corner / HIL	Mode transitions; stop curve; fault injection; REQ-M25 path
G4	Optional mule	ISO 2631-1 and handling on donor chassis
G5	Licence / transaction readiness	Buyer-accepted package; commercial term sheet; IP filing status; ownership chain

13.2 Phase schedule (planning; no commitment without funding).

Phase	Month	Content
Ph0	0–2	CAD envelope, lock faces, ICD-001 draft; procurement package after ICD baseline
Ph0b	2–6	Supplier lead time (4–8 months planned); COTS surrogate bench; SIM-001 numerical run; 90-day Option deliverables limited to SIM-001 + ICD-001 draft
Ph1	6–10	Single-tower force, thermal, power-loss, lock cycle; ACT-001 map start

Phase	Month	Content
Ph2	10-15	Four-corner 6-DOF; mode transitions; stop curve; power-loss from each mode; REQ-M25 path
Ph3	15-21 (optional)	Donor-chassis ISO 2631-1 and handling evaluation

Schedule assumes a funded programme. Without funding, dates are not commitments. Option deliverables under a 90-day window are limited to SIM-001 and ICD-001 draft.

13.3 Risk register (likelihood × impact on a 1-5 scale).

ID	Risk	L	I	L×I	Owner
R-01	Core tower mass exceeds limit at PARAM-MASS-CONFIDENCE	4	4	16	Module
R-02	Km not achieved / not demonstrated by measured map	3	5	15	Actuator
R-03	Lorentz thermal or force-gap limits	3	5	15	Actuator
R-04	Macro-micro clash or carriage failure	3	4	12	Mechanical
R-05	Stroke consumed without leveling	3	4	12	Hydraulics
R-06	Regulatory path unclear	3	4	12	Homologation
R-07	Vehicle P90 exceeds host mass ceiling	4	4	16	Mass
R-08	Per-tower crash load vs capture frame / stops / hardpoints	4	5	20	Structures
R-09	Comfort unproven until L1/L2 evidence	4	4	16	Sim + PoC
R-10	EMI / field-exposure cost	2	4	8	EMC
R-11	Hydraulic mass above STUDY	3	3	9	Hydraulics
R-12	Liability if safety case incomplete	2	5	10	Legal + safety
R-13	SPORT lock reliability / partial engage	3	4	12	Lock
R-14	Mode arbitration false positive / negative	3	3	9	Controls
R-15	Horizontal restraint undersized (REQ-M25)	3	5	15	Structures
R-16	Clearance-stack energy under-estimated	3	4	12	Structures
R-17	Unintended SPORT lock at speed on rough road	3	4	12	Systems
R-18	Accumulator pre-charge loss	3	3	9	Hydraulics
R-19	Hydraulic leak / fire	2	5	10	Hydraulics
R-20	Common-cause sensor fault	3	4	12	Systems
R-21	Magnet demagnetisation	2	4	8	ACT-001
R-22	Packaging residual remains negative at G1	4	4	16	Packaging

13.4 Open Items Register (complete — replaces the fragmented challenge list of R5.16).

#	Open item	Owner path	Target gate
1	Packaging residual versus LOCKED language — residual plan to ≥0	\$4 / C-01 / CFG-001	G1
2	REQ-M07 quantitative settle limits frozen	\$7 / SAF-001	G2
3	REQ-M15 operability under dual failure (power + hydraulic)	\$7 / physical trial	G3
4	REQ-M21 single-sensor and disagreement behaviour	\$7 / SIM-HIL	G2
5	REQ-M25 numeric free play, stiffness and load share	\$7 / STR-001 / ICD-001	G2
6	Progressive-stop force-displacement curve and design energy event	STR-001	G3
7	IP ownership chain completeness and provisional filing	IP-001 / counsel	Before broad T1
8	Configuration discipline — one value, one place under CFG-001	CFG-001	G1
9	State-machine verification (S3 entry, S4 exit, S5 trigger class)	\$8 / HIL	G2
10	Km evidence chain and gap tolerance envelope	ACT-001	G2
11	48 V total system current budget	ENE-001	G2
12	Tower mass statistical basis (P90 confidence)	Supplier data / CFG-001	G1

13.5 Immediate work packages.

Priority	Work package	Closure deliverable
G1	Freeze CAD envelope, lock faces, leveling, fail-closed valves; residual plan	3-D package + clash report
G1	ICD-001 draft (mech / elec / comm / safety / lock)	Buyer-readable ICD set
G1	SIM-001 assumptions + first COMFORT vs passive run	Plot pack v0.1 + configuration hash
G2	Single-tower force, thermal, power-loss, lock cycle	Bench data package
G2	Formal HARA + FTA for capture and asymmetric lock	SAF-001 work products
G2	STR-001 stop curve + REQ-M25 horizontal loads	Structures memo
G3	Four-corner rig mode transitions	6-DOF + ≤ 3 s records
G5	Path A Option / term-sheet support	Commercial pack under counsel (COM-001)

14. HOST VARIANTS (PATH B) — ENGAGEMENT TARGETS

Path B host figures are STUDY for vehicle programmes. Module Path A does not require the buyer to adopt host powertrain or GVW figures. Sole valid host budgets below use a unified 100 kg occupant policy. Engine descriptions use class descriptors only (no brand names). Geometry LOCKED values remain engagement targets and are subject to the Packaging Feasibility Gate in §4.

14.1 Geometry lock — both variants.

Parameter	V-L Urban Light	V-H Military / Off-road	Status
Overall length	5,000 mm	5,500 mm	LOCKED
Overall width (mirrors excluded)	2,000 mm	2,200 mm	LOCKED
Overall height (SPORT / seated)	1,700 mm	2,000 mm	LOCKED
Wheelbase	3,000 mm	3,300 mm (front to mid of rear tandem)	LOCKED
Ground clearance (SPORT)	200 mm	300 mm	LOCKED
Wheels / drive	4×4 (four wheels)	6×6 — front axle + dual rear tandem	LOCKED
Design occupants	4	6	LOCKED
Passenger cell	Full CFRP structural cell	Full CFRP structural cell	BASELINE
Roll structure	FIA-referenced CFRP/hybrid	FIA-referenced CFRP/hybrid	Reference only — no homologation claim
GVW hard ceiling	4,000 kg	7,000 kg	LOCKED

14.2 Powertrain lock — hybrid ICE + electric assist (class descriptors).

Parameter	V-L	V-H	Status
Architecture	Hybrid (ICE + e-drive)	Hybrid (ICE + e-drive)	LOCKED
ICE class	V8, 4.0 L twin-turbo class	V12, 6.0 L twin-turbo class	LOCKED class
ICE power (planning)	≈430–450 kW class	≈460–500 kW class	TARGET
Electric traction (planning)	≈150–180 kW total	≈200–250 kW total	TARGET
Battery usable energy	20 kWh	40 kWh	LOCKED
Fuel tank	80 L	150 L	LOCKED
Combined range target	1,000 km (STUDY) pending road-load at 90/120 km/h	700–800 km (STUDY) pending road-load	STUDY

Both range figures are STUDY pending road-load calculation at 90/120 km/h under ENE-001. V-L 1,000 km assumes an optimistic combined intensity and shall be recomputed before any external claim.

14.3 Closed mass budget — V-L (sole valid host budget).

Occupants: 4 × 100 kg = 400 kg. Luggage allowance planning 100 kg. Illustrative GVW design ≈ 3,365 kg class. Hard ceiling 4,000 kg.

WBS line	kg	Notes
CFRP passenger cell	280	4-seat hard-points
FIA-referenced roll members	45	
Closures + glass + panels	230	
Interior 4-seat + HVAC + cabin LV	200	
Lower chassis	340	4×4
V8, 4.0 L twin-turbo class ICE + ancillaries	275	Hybrid ICE
Hybrid transmission / coupling	160	

WBS line	kg	Notes
E-motors + inverters	130	
Battery 20 kWh usable	145	LOCKED energy
HV/LV + cooling	95	
Axles / steering	260	
Wheels + tyres + brakes	210	
Isolation 4× class-S corners	280	CMFSB
Fuel tank 80 L + reference fuel	75	
Exhaust + fluids	70	
Programme reserve	70	
Curb (no occupants)	≈2,865	Line sum
Occupants 4 × 100 kg	400	Policy
Urban cargo	100	
GVW design case	≈3,365	Margin to 4,000
Hard ceiling	4,000	LOCKED

14.4 Closed mass budget — V-H (sole valid host budget).

Occupants: 6 × 100 kg = 600 kg. Luggage allowance planning 400 kg. Illustrative GVW design ≈ 6,150 kg class. Hard ceiling 7,000 kg operational.

WBS line	kg	Notes
CFRP passenger cell	420	6-seat hard-points
FIA-referenced roll members	70	
Closures + glass + reinforced exterior	400	
Interior 6-seat + HVAC + LV	330	
Lower chassis reinforced	720	6×6 structure class
V12, 6.0 L twin-turbo class ICE + ancillaries	390	Hybrid ICE
Hybrid transmission / coupling	190	
E-motors + inverters	180	
Battery 40 kWh usable	260	LOCKED energy
HV/LV + heavy cooling	160	
Three-axle 6×6 driveline	580	Front + dual rear
Wheels + off-road tyres + brakes	480	Six corners
Isolation 4× class-M/L corners	380	CMFSB
Light protection package (optional)	200	Else → reserve
Fuel tank 150 L + reference fuel	140	
Fluids	100	
Programme reserve	150	
Curb (no occupants)	≈5,150	Line sum with 40 kWh
Occupants 6 × 100 kg	600	Policy
Mission payload allowance	400	
GVW design case	≈6,150	Margin to 7,000
Hard ceiling	7,000	LOCKED

14.5 Host requirements additions.

ID	Requirement	Status
REQ-V01	V-L GVW claims shall remain ≤4,000 kg including stated occupant policy	TARGET
REQ-V02	V-H GVW claims shall remain ≤7,000 kg including stated occupant/payload policy	TARGET
REQ-V03	Both variants implement CFRP passenger cell within §14 mass budgets	TARGET
REQ-V04	Both variants implement FIA-referenced roll structure; no homologation claim without formal process	Ongoing
REQ-V08	V-L ICE: V8, 4.0 L twin-turbo class	LOCKED class
REQ-V09	V-H ICE: V12, 6.0 L twin-turbo class	LOCKED class

ID	Requirement	Status
REQ-V10	Range: V-L 1,000 km (STUDY); V-H 700–800 km (STUDY) — both pending road-load at 90/120 km/h	STUDY

15. SUPPLIER AND ICD OUTLINE

15.1 Supplier questionnaire themes.

Domain	Questions the supplier must answer
Hydraulic macro	Cylinder force at pressure; seal life; fail-closed valve response time; orifice-hold option capability
Lorentz / actuator	Km measured across gap tolerance; continuous and peak force; thermal map; mass; demagnetisation limits
Lock	Engage/release time under load; status sensing; shear capacity; contamination tolerance
Stops	Force-displacement curve; energy rating; temperature sensitivity; inspection interval
Sensors	Stroke, force, pressure, temperature accuracy; fault behaviour; independence from control sensors
Electronics	48 V drive capability; current limit; diagnostic coverage; EMC class

15.2 ICD-001 minimum chapter list. Hardpoint geometry and loads; hydraulic ports and fluid specification; electrical power and signals; mode and interlock logic summary; progressive-stop curve reference; thermal limits; installation torque and sequence; maintenance and rescue release; EMC and grounding; configuration identity and change control.

15.3 Configuration management. All TARGET and STUDY values that become design inputs shall be configuration-managed under CFG-001. DOC-00 remains the narrative baseline; freeze occurs in the ICD and part drawings, not by DOC-00 revision alone. One value, one place: conflicting numbers between sections are defects and shall be corrected at the next revision.

16. TRANSACTION READINESS (PATHS A, B, C)

16.1 Path A (module licence) — primary commercial route. Field licence or supply of ATFSV-MOD-001 without requiring the buyer to adopt host powertrain or GVW figures. Detailed commercial terms (Option validity, exclusivity, fees, credit) are counsel-controlled T3 content in COM-001. T1 retains only this one-line reference: Path A Option is available under COM-001; non-exclusive evaluation by default; limited exclusivity as a paid add-on.

16.2 Path B (vehicle programme). Uses the locked hosts in §14 when the buyer funds a vehicle path. Not required for Path A.

16.3 Path C (IP assignment). Price and conditions are bilateral, subject to counsel, patent filing status and evidence gates. Contracting party: Amirbahador Atai (individual) until formal assignment to a legal entity.

16.4 DOC-00 boundary statement. DOC-00 is not a certificate, type approval, warranty, patent grant or price list. Contradicting external presentations shall be withdrawn. DOC-00 alone does not constitute an offer of exclusive sale.

16.5 Intellectual property filing status (Defect 7 closure — honest buyer posture).

Status as of issue date. No patent application has been filed as of 2026-10-06 per project record. When a provisional application is filed, this section shall be updated to state: Provisional application No. [number], [office], [date]. Internal distribution rule (not a buyer obligation): prefer provisional filing with patent counsel before further broad T1 distribution to third parties. NDA alone may be insufficient in absolute-novelty jurisdictions. No novelty, enforceability or FTO conclusion is asserted by this document.

16.6 Conditions precedent to any assignment or exclusive licence (checklist).

#	Condition	Owner
1	Provisional or full filing record in force	Counsel / IP-001
2	Invention disclosure and inventor chronology complete	IP-001
3	Prior-art and landscape review recorded	Counsel
4	Ownership chain and assignment path defined	Counsel / COM-001
5	NDA and disclosure-history register current	IP-001
6	Evidence gates agreed (e.g. ICD + SIM or bench plan)	Technical + commercial
7	FTO opinion path defined (may be a post-signing condition)	Counsel
8	Entity assignment path defined if seller remains an individual	IP-001 / COM-001

16.7 What Path A delivers (planning summary — full terms in COM-001).

Typically included under a closed Path A licence	Not included unless separate agreement
Rights to use defined CMFSB module architecture in a named field of use	Worldwide assignment of all related IP
Controlled package: ICD, datasheet, safety behaviour, dual-mode rules	Turnkey certified vehicle or type approval
Agreed engineering support hours	Unlimited R&D for the buyer's entire vehicle line

Typically included under a closed Path A licence	Not included unless separate agreement
Updates within a defined revision window if contracted	Guaranteed comfort percentage without L1/L2 evidence

17. GOVERNANCE AND SIGN-OFF

17.1 Master Document Register (single source of IDs).

ID	Title / scope	Status at R5.18
DOC-00	This baseline	R5.18
MOD-001	Module datasheet (1–3 page buyer summary)	To issue from DOC-00
ICD-001	Interface control document	OPEN / draft under Option
SIM-001	Multi-band simulation	OPEN
STR-001	Structural / progressive-stop / crash-path	OPEN
LCK-001	Lock system	OPEN
ACT-001	Actuator thermal, force, efficiency, gap map	OPEN
HYD-001	Hydraulic macro stage, valves, leveling	OPEN
SAF-001	HARA / FMEA formal pack	OPEN
TST-001/002	L2 / L3 test plans	OPEN
CFG-001	Configuration management	OPEN
ENE-001	Energy / power budget detail	OPEN
BRK-001	Brake / horizontal load interface	OPEN
REG-001	Regulatory pathway	OPEN
IP-001	IP / patent filing (counsel)	OPEN
COM-001	Commercial Annex (T3 / counsel)	OPEN

17.2 Priority before serious OEM or investor engagement. (1) ACT-001, STR-001 and SIM-001 evidence; (2) SAF-001 including quantitative power-loss limits, sensor independence and horizontal load path; (3) IP-001 filing, prior art, claim chart and ownership chain; (4) packaging residual plan to non-negative under configuration control.

17.3 Release and sign-off.

Role	Name / status	Date / target
Legal Owner / Author	Amirbahador Atai	2026-10-06
Independent Technical Reviewer	TBD — to be appointed by PARAM-REVIEWER-DEADLINE	Earlier of 2026-12-05 and G1
Approval Authority	TBD after independent review	After G1 evidence
Documentation authority	ATFSV Project Engineering Documentation Authority	2026-10-06

Status: CONTROLLED / Contract Diligence Baseline. Not approved for road use, production or certification credit. All TARGET, STUDY, ASSUMPTION, DERIVED, LOCKED and OPTION values require formal verification before design release.

ANNEX T3-A — IP SCHEDULE (T3 — COUNSEL CONTROLLED — REMOVE FROM ALL T1/T2 DISTRIBUTION)

T3-A.1 Proposed patent families (for counsel review only).

Family	Scope	Treatment
A	Core mechanically captured macro-micro isolation architecture	Patent candidate
B	Hybrid hydraulic macro + high-bandwidth micro actuator architecture	Patent candidate
C	Positive SPORT lock and mechanical capture geometry	Patent candidate
D	Safety state machine; S4 passive-support with quantitative envelope; transition constraints	Patent / trade-secret review
E	Dynamic mode arbitration and control; REQ-M21 interlock logic	Patent / software-protection review
F	Progressive-stop and independent crash path with force-displacement absorption	Patent candidate
G	Modular host-class integration interface (four-tower ICD, load sharing)	Patent candidate
Trade secrets	Actuator tuning, calibration maps, control parameters, manufacturing processes, supplier-specific implementation	Trade-secret candidate

T3-A.2 Invention-disclosure candidates (subject to counsel): (1) mechanically captured macro-micro isolation architecture; (2) passive mechanical support combined with high-bandwidth Lorentz micro-actuation; (3) dual-mode COMFORT/SPORT with positive lock and capture invariants; (4) progressive-stop independent crash path; (5) host-class adaptable four-tower ICD. No attitude / pyramidal underbody family is asserted in this baseline.

T3-A.3 Filing and ownership record (to be completed and dated by counsel).

Item	Record
Provisional application(s)	None filed as of 2026-10-06 per project record. Update: No. [___], Office [___], Date [___]
Inventor(s)	Amirbahador Atai [confirm complete list and chronology]
Current owner	Amirbahador Atai (individual)
Assignment to entity	[Entity, date, instrument — TBD]
Contractor / supplier IP clauses	[TBD]
Confidentiality and disclosure history	[NDA register — TBD]
Jurisdiction strategy	[TBD by counsel]
FTO opinion	[TBD by counsel]

IP-001 content shall include: invention disclosure, inventor chronology, prior-art search, patent landscape, claim chart, potentially conflicting families, jurisdiction strategy, ownership chain, assignment records, confidentiality history, FTO review.

T3-A.4 Term-sheet topic schedule (topics only; values are bilateral and live in COM-001). Field of use · territory · exclusivity and its fee · Option period and exercise mechanics · milestone and royalty structure · improvements and grant-back · ownership and title warranties · IP representations and indemnity · data-room and escrow arrangements · supplier flow-down · audit rights · export-control compliance · confidentiality and publication · termination and reversion · governing law and dispute resolution.

ANNEX B — REFERENCES AND GLOSSARY

Reference	Use	Status
ISO 26262	Functional-safety process for E/E elements	Candidate; compliance not claimed
ISO 21448	Safety of the intended functionality (SOTIF)	Candidate
ISO 2631-1	Whole-body vibration evaluation	Candidate
ISO 8608	Road-surface profiles for simulation	Candidate
ISO 4413	Hydraulic fluid power — general rules and safety (REQ-M24)	Reference
ISO 13849-1	Lock and valve safety functions if used	Candidate
UN R100 / ISO 6469-3	HV safety if applicable	Candidate
UN R94 / R95 / R14 / R16	Crash and restraint matters if applicable	Candidate
EN 1789 / EN 1865	Ambulance planning references (Application A only; no clinical claim)	Candidate
FIA Appendix J	Roll-structure philosophy only (verify article applicability)	Reference

Glossary. CMFSB — Captured Mechanical Floating Safety Body. S4 — power-loss passive-support state. COMFORT / SPORT — dual running-gear modes (states S2 / S2S). Application A — ambulance and clinical-transport planning context (no clinical outcome claimed). Path A / B / C — module licence, vehicle programme, IP assignment. VDV — vibration dose value. MSDV — motion sickness dose value. P50 / P90 — median and 90th-percentile planning values. LOCKED — programme engagement decision; not packaging closure and not test verification.

END OF ATFSV-DOC-00 | REVISION 5.18

Single content layer. No duplicated body. No blank pages. No attitude/pyramid option in this baseline. Seven document-control defects from the R5.16/R5.17 review are closed at the document level. All TARGET, STUDY, ASSUMPTION, DERIVED, LOCKED and OPTION values require formal verification before design release. Factual change log is maintained separately from this baseline.

ANNEX C — CONTROLLED FIGURE DESCRIPTIONS AND EVIDENCE RECORDS

C.1 Controlled schematic set. The following figures are configuration-controlled under CFG-001. Textual descriptions below define minimum content. CAD models and drawings supersede narrative where conflict exists after ICD freeze.

Figure	Title	Minimum content	Owner
Fig. 1	System context	Upper body, four towers, host hardpoints, layer stack top→bottom (cell → micro → macro → chassis)	Systems
Fig. 2	Corner force path	Macro cylinder, Lorentz carriage, progressive stop, SPORT lock; vertical and horizontal load arrows	Structures
Fig. 3	Mode state graph	S0-S8 with transition arrows; REQ-M21 inhibit on SPORT→COMFORT; power-loss to S4 from any state	Systems
Fig. 4	Power-loss timing	Command loss → valve close → settle within envelope; lock fail-locked if engaged; time axes OPEN until test	Safety
Fig. 5	Packaging stack (V-L)	Overall height, GC, chassis, tower, travel, cabin ≥1,000 mm from H-point, roof; residual call-out	Packaging
Fig. 6	Lock seat band	Seat faces, pin/claw geometry class, status-sensor locations, clearances both modes	LCK-001

C.2 Evidence record requirements (every L1-L3 result).

Field	Requirement
Software / parameter set	Version identifier and configuration hash
Hardware configuration	Tower serial or surrogate ID; lock fitment; sensor set
Road / profile ID	ISO 8608 class or measured profile reference
Payload / mass state	Empty / occupied / maximum as §6
Mode	COMFORT (S2), SPORT (S2S), or transition case
Reference passive k/c	Frozen values stated on every comparative plot
Time histories	Force, stroke, acceleration, lock status as applicable
Uncertainty	Method stated where numerical claims are made
Review status	PRELIMINARY until independent review disposition

C.3 Frequency, deflection and leveling (engineering note).

Corner mass share for the occupied reference ($\approx 2,650 \text{ kg} / 4$) $\approx 662.5 \text{ kg}$. At $k \approx 40 \text{ kN/m}$ about the leveled operating point, $f_n \approx 1.23 \text{ Hz}$. The 40 kN/m figure is the effective rate about the leveled operating point; static sag is taken by gas-spring preload and active leveling (REQ-M22), not by the $\pm 50 \text{ mm}$ dynamic stroke. A payload change $\Delta F \approx 2.4 \text{ kN}$ would give $\approx 60 \text{ mm}$ at 40 kN/m ; therefore active leveling is required to preserve dynamic stroke for isolation. Target isolation band remains STUDY pending SIM-001. Full load-deflection curve shall be frozen in the ICD.

C.4 Clearance and stroke stack (nominal → stop).

Segment	Planning description	Example sensitivity
Nominal height	Leveling set-point under stated payload	Shifts with payload if leveling fails
Macro envelope	$\pm 50 \text{ mm}$ body-relative about nominal	100 mm total range
Pre-trigger / free play to stop	Gap after macro limit before stop energy rise	If $s=30 \text{ mm}$ and $a \approx 20 \text{ g}$, $v=\sqrt{(2as)} \approx 3.43 \text{ m/s}$
Stop stroke	Energy absorption displacement (STR-001)	Curve OPEN — not a compliance claim

Stop energy illustration (occupied upper body $\approx 2,650 \text{ kg}$, $v \approx 3.43 \text{ m/s}$): $E \approx \frac{1}{2} \cdot m \cdot v^2 \approx 15.6 \text{ kJ}$ ($g = 9.81$). Until STR-001 freezes the force-displacement curve, all energy figures remain STUDY. S5 (impact preparation) may reduce effective free play by pre-positioning; energy and velocity numbers must be recomputed for the frozen S5 geometry.

C.5 Integration notes for 4×4 and 6×6 host vehicles.

Topic	4×4	6×6
Module interface	Four towers at upper-body hardpoints	Same four towers if load class held
Integration risk	Lower — preferred pilot path	Higher (axle, steer, packaging, GC)
First-contract bias	Preferred for road / limited-series pilot	Second SKU or buyer-funded utility path
Differentiation	Road / limited series / ambulance pilot	Extreme / specialty / defence-light

Module hardpoints must match host chassis stiffness and packaging. Ground clearance and overall height are LOCKED engagement targets in §14 and remain subject to the Packaging Feasibility Gate (§4). Brake and propulsion torque reactions must not overload horizontal links (REQ-M25). Hybrid powertrain packaging is a host responsibility; the module ICD defines only interface loads, power and signals.

C.6 Readiness matrix for external engagement.

Use case	R5.18 suitable?	Condition
NDA + T1 technical diligence	Yes	This document under NDA
Paid Option negotiation	Yes	Bilateral term sheet; fee not published here (COM-001)
Module licence term sheet	Conditional	Prefer ICD draft + SIM or bench plan committed
Large upfront assignment / exclusive sale	Not on DOC-00 alone	Requires IP filing status, FTO counsel, evidence
Production / type-approval claim	No	Explicit non-claim
Advertising measured ride comfort	No	Until L1/L2 configuration-managed results

C.7 Residual OPEN items that do not block NDA (but block exaggerated claims). OPEN: H-point stack residual freeze; STR-001 stop curve; continuous RMS thermal evidence; provisional patent filing number; supplier P90 mass; ASIL assignment; REQ-M07 quantitative settle limits; REQ-M25 numeric free play. These are tracked in §13.4 and the risk register. Presenting any of them as closed is a communication defect.

ANNEX D — ACCEPTANCE CRITERIA, NON-CLAIMS COMMUNICATION AND CHANGE DISCIPLINE

D.1 Path A Option deliverable acceptance (90-day window — planning).

Deliverable	Acceptance criterion (planning)	Not accepted if
ICD-001 draft	Buyer-readable chapters covering hardpoints, power, signals, mode logic, stops reference	Missing safety or lock chapters
SIM-001 run pack	COMFORT vs frozen passive on ≥ 1 rough ISO 8608 profile; hash recorded	No passive reference or no hash
Configuration index	Document IDs and revision status for all artefacts in the Option pack	Conflicting numbers across files

Option acceptance does not imply production readiness, comfort percentage, or certification credit. It confirms that the agreed planning artefacts were delivered under configuration control.

D.2 External communication rules (binding on sales and diligence materials).

Allowed	Not allowed
Describe architecture: captured body, COMFORT/SPORT, macro/micro, progressive stops	Invent measured comfort percentages
State dual-mode behaviour and safety invariants in qualitative terms	Claim type approval, ASIL, or crash certification
Cite TARGET / STUDY / ASSUMPTION labels with their status vocabulary	Present STUDY or ASSUMPTION as verified performance
Reference this DOC-00 under NDA for diligence	Publish fixed licence prices from DOC-00
State patent filing status honestly (none filed until updated)	Imply novelty, grant, or FTO from DOC-00 alone

D.3 Change discipline (one value, one place).

When a numeric value is frozen, it shall appear in exactly one authoritative location: either the Master Parameter Register (§3) or a named downstream document (ICD, STR-001, ACT-001). Narrative sections may reference the parameter ID; they shall not restate a conflicting number. If a conflict is discovered, the downstream configuration-controlled source wins after G1, and DOC-00 shall be revised at the next issue. Historical tables retained for traceability shall be labelled HISTORICAL or SUPERSEDED.

D.4 Buyer diligence checklist (recommended questions).

#	Question for the selling party	Where answered in R5.18
1	What is locked versus still STUDY/OPEN?	§1.6 vocabulary; §3 register; §13.4 open items
2	Is packaging closed for the quoted overall height?	§4 — No; residual explicit
3	What is the patent filing status?	§16.5 — None filed as of issue date
4	What does Path A include and exclude?	§16.1 and §16.7
5	What evidence exists at L1/L2 today?	§12 — ladder; currently L0 architecture
6	Who is the contracting party?	Cover and §16.3 — Amirbahador Atai (individual)
7	How are commercial terms controlled?	COM-001 (T3); not published in T1

D.5 Relationship of this baseline to earlier numerical packages.

Earlier issues (R5.11–R5.17) contained evolving mass, floor, range and packaging figures. R5.18 withdraws those files as active design inputs. Where a prior number is needed for audit, the project Change Log and configuration archive retain it under SUPERSEDED labelling. Buyers shall not mix numbers from withdrawn files with R5.18 parameters.

D.6 Summary of document-control posture for a real buyer.

Topic	R5.18 posture
Content integrity	Single layer; no duplicated body; no blank pages
Engineering honesty	OPEN items listed; residual packaging negative and explicit
Commercial boundary	No prices; COM-001 for terms; Path C available bilaterally
IP honesty	None filed; provisional preferred before broad T1
Attitude / lean option	Not part of this baseline; separate option if pursued later
Certification	None claimed; TRL 2–3 stated on every page footer

END OF ANNEX D · ATFSV-DOC-00-R5.18 complete

This annex completes the buyer-facing control package. Core sections 0–17, Annex T3-A, Annex B, Annex C and Annex D together form the single controlled baseline. No further body content follows.

ANNEX E — MASTER POSITION STATEMENT AND DOCUMENT MAP

E.1 Master position statement.

R5.18 is the Contract Diligence Baseline for ATFSV-MOD-001. It defines a mechanically captured macro-micro isolation architecture with COMFORT and SPORT modes, owned requirements REQ-M01 through REQ-M25, the SIM-001 verification skeleton, gates G0-G5, phase schedule Ph0-Ph3 (with Ph0b), risk register R-01 through R-22, a module-visible mass table, host WBS budgets for V-L and V-H under a unified 100 kg occupant policy, horizontal-restraint requirements, an explicit packaging residual (V-L ≈ −120 mm OPEN), value KPIs without invented comfort percentages, and commercial Path A engagement without published fixed licence prices. TRL remains 2-3. Legal owner is Amirbahador Atai. No test result, certification, patentability or FTO is asserted. The remaining work is evidence and formalisation: execute SIM-001, run single-tower bench, freeze ICD and STR-001, close packaging residual under CFG-001, appoint the independent reviewer, and pursue counsel-led IP filings.

E.2 Document map (what lives where).

Section / Annex	Primary content
\$0	Revision control; seven defect closures; C-register
\$1	Purpose, non-claims, value, disclosure tiers, status vocabulary
\$2	Architecture, safety invariants, modes, hold, horizontal restraint
\$3	Master Parameter Register (sole active numeric source with \$7)
\$4	Packaging Feasibility Gate; LOCKED ≠ closed; residual
\$5	Power and current envelope; copper vs system power
\$6	Mass, static floors, per-class load cases
\$7	REQ-M01-REQ-M25
\$8	State machine S0-S8 and complete transition table
\$9	FMEA and preliminary HARA
\$10	Actuator map, gap, thermal
\$11	Mode sequences and lock architecture
\$12	Verification framework and evidence ladder
\$13	Gates, phases, risks, open items, work packages
\$14	Host variants V-L / V-H (Path B)
\$15	Supplier and ICD outline
\$16	Transaction readiness Paths A/B/C; patent status
\$17	Governance, document register, sign-off
Annex T3-A	IP schedule (counsel controlled — strip from T1/T2)
Annex B	References and glossary
Annex C	Figure descriptions, evidence records, integration notes
Annex D	Acceptance criteria, communication rules, buyer checklist
Annex E	Master position statement and document map

E.3 Single-layer integrity statement.

This file contains exactly one continuous narrative from cover through Annex E. There is no appended corrections block, no duplicated body, and no blank page. Footer page numbers are sequential. If any external copy shows duplicated sections or conflicting revision IDs, that copy is not the controlled R5.18 issue and shall be withdrawn.

END OF ATFSV-DOC-00 | REVISION 5.18

Single content layer. Seven document-control defects closed. No attitude/pyramid option in this baseline. All TARGET, STUDY, ASSUMPTION, DERIVED, LOCKED and OPTION values require formal verification before design release. Factual change log maintained separately from this baseline.